# Substitution

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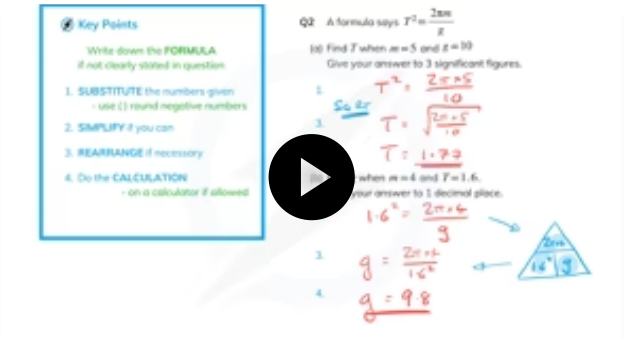


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## Substitution

### What is substitution?

- **Substitution** means **replacing** a letter (variable) in a formula with a given **number**

### How do I substitute numbers into a formula?

- Write down the **formula**
- **Replace** (substitute) the letters in the formula with the given numbers
  - If substituting in a **negative number**, it is important to put **brackets** around it
    - For example, (-3)
- **Simplify** any numerical calculations
- **Calculate** the final value
- Sometimes the result is an **equation** which you can then **solve**

Help me understand

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

### Examiner Tips and Tricks

Be careful with negatives and powers. For example, if $x = -4$ then $x^2 = (-4)^2 = 16$.

## Worked Example

(a) Given that $x = 2$ and $y = -4$, find the value of the expression:

(i) $2x(x + 3y)$

(ii) $y^2 + y^3$

**Answer:**

**(i)**

Substitute the numbers given
Use brackets () around negative numbers

$$2 \times 2 \times (2 + 3 \times (-4))$$

Complete the calculation
Show every step of working, following the order of operations correctly

$$= 2 \times 2 \times (2 - 12)$$
$$= 2 \times 2 \times (-10)$$
$$= 4 \times (-10)$$

**-40**

**(ii)**

Substitute the numbers given
Use brackets () around negative numbers

$$(-4)^2 + (-4)^3$$

Work out the powers remembering to include the brackets

$$16 + (-64)$$

Complete the calculation

$$16 - 64$$

**-48**

(b) The formula $P = 2l + 2w$ is used to find the perimeter, $P$, of a rectangle of length $l$ and width $w$.

Given that the rectangle has a perimeter of 20 cm and a width of 4 cm, find its length.

**Answer:**

Substitute the values you are given for $P$ and $w$ into the formula

$$20 = 2 \times l + 2 \times 4$$

Simplify

$$20 = 2l + 8$$

$12 = 2l$

Divide both sides by 2

$l = 6 \text{ cm}$

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